

*Manta birostris*

Giant manta ray



**Length** Up to 23 ft (7 m)  
**Weight** Up to 2 tons (1.8 tonnes)  
**Breeding** Viviparous  
**Status** Vulnerable

**Location** Tropical and sometimes warm temperate waters worldwide

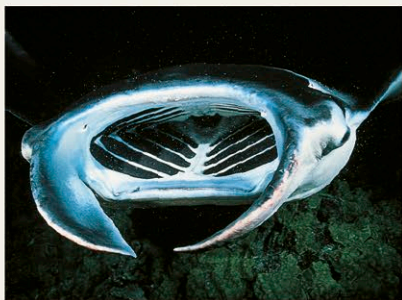


Measuring up to 23ft (7m) across, the manta—also known as the devil ray—is the world’s largest ray, swimming through the open ocean like a large bird on slowly flapping wings. However, despite its sinister look, it is a harmless plankton-eater, filtering food with its gills. Its mouth, unlike those of other rays, is positioned in front of its body, allowing it to feed continuously as it moves along. Normally a leisurely swimmer, it can suddenly accelerate if threatened, and can even leap out of the water, possibly to avoid predators. Mantas are usually solitary, although they sometimes swim in small, loosely organized schools. They prefer warm waters, and are mostly found in the tropics. In 2009, the smaller coastal mantas often seen by divers were found to

be a distinct species named the reef manta ray (*Manta alfredi*). During the breeding season, the male chases the female, swimming beneath her so that their undersides face each other, and inserts one of his claspers. The female usually gives birth to one or 2 young per year, each about 4ft (1.2m) wide.

FUNNELING FOOD

Giant manta rays have large, flaplike cephalic lobes, and they use these to funnel prey into their mouths. The water passes out through the ray’s gills, but the prey is trapped on transverse spongy plates that bridge the gaps between the gill bars. Mantas eat small, schooling fishes, as well as smaller planktonic animals.



triangular pectoral fins

"wingspan" over 21 ft (6.5 m)

flaplike cephalic lobes

UNDERWATER FLIGHT

The manta has large, triangular pectoral fins with pointed tips, and a short tail without a fin or sting. In normal circumstances, its fins beat up and down about once every 4 or 5 seconds, but the manta can accelerate rapidly, sometimes somersaulting out of the water and crashing back onto the surface with a massive impact.

*Rhinobatos productus*

Shovelnose guitarfish



**Length** Up to 5 ft (1.5 m)  
**Weight** 33–40 lb (15–18 kg)  
**Breeding** Viviparous  
**Status** Near threatened

**Location** E. Pacific



Named for its unusual shape, the shovelnose guitarfish has a broad head with a clear, cartilaginous area on either side of the snout. Its pectoral fins are wide, like

those of a typical ray, but the rest of its body is cylindrical, like that of a shark. Found in the warmer areas of the eastern Pacific Ocean, it tends to move inshore in the summer months. The shovelnose guitarfish is generally solitary but sometimes gathers in large numbers for breeding. Found in shallow water around beaches, bays, estuaries, and reefs, it lies partially buried in sand

or mud. It feeds at night on small bottom-dwelling fish, clams, crustaceans, and worms, which it uncovers from the seabed. The female produces 5–25 young, which are about 6in (15 cm) long at birth.

cylindrical, sharklike body

triangular tail fin

rounded pectoral fins

*Urolophus halleri*

Round ray



**Length** 20–23½ in (50–60 cm)  
**Weight** 4½–8¾ lb (2–4 kg)  
**Breeding** Viviparous  
**Status** Least concern

**Location** E. North Pacific



Unlike typical stingrays, the round ray and its relatives, sometimes known as stingarees, have an almost circular body. The body is very flattened, and the tail is relatively short, with a rounded fin at its tip. During warm seasons, large numbers sometimes move into shallow beaches where they may sting bathers who are unfortunate enough to step on them. As with other stingrays, the sting of this can be intensely painful.

sting partway along tail

tail fin

*Taeniura lymma*

Blue-spotted stingray



**Length** Up to 6½ ft (2 m)  
**Weight** Up to 66 lb (30 kg)  
**Breeding** Viviparous  
**Status** Near threatened

**Location** Indo-Pacific



One of the most handsome members of the stingray family, this species has blue spots scattered over its body. In addition, it has blue stripes along the side of its tail and a lighter underside. Like all stingrays, this fish has a toxic spine located at the base of its tail and is capable of inflicting a nasty sting if stepped on or

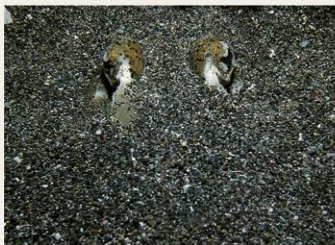
mishandled. Although mainly diurnal, it is sometimes active at night. It moves into shallow areas with the advancing tide to feed on small fishes and invertebrates, especially crustaceans and worms. It is often found in sandy areas next to reefs, and rests in reef caves and crevices when not feeding. The female gives birth to 3–7 young and, like the South American freshwater stingray (see opposite), she nourishes her young by secreting uterine “milk.” The colorful appearance of the blue-spotted stingray has made it a popular fish in the aquarium trade and a favorite subject of underwater photographers.

round body  
blue spots on body

small pelvic fins

stripes on tail

LYING IN WAIT



Blue-spotted stingrays normally actively forage for food, but they rest with their bodies largely hidden by mud or sand, with just their eyes projecting above the surface. In shallow water, hidden rays are easily stepped on.

